Commands :

1. type

```
phanindra1815@Phanindras-PC:~$ type cd
cd is a shell builtin
phanindra1815@Phanindras-PC:~$ type ls
ls is aliased to `ls --color=auto'
phanindra1815@Phanindras-PC:~$
```

2. ulimit

```
phanindra1815@Phanindras-PC:~$ ulimit -n
10240
phanindra1815@Phanindras-PC:~$
```

3. nmap

```
phanindra1815@Phanindras-PC:~$ nmap -p 22,80 localhost
Starting Nmap 7.993 ( https://nmap.org ) at 2026-10-29 09:39 +0000
Warning: Hostname localhost resolves to 2 IPs. Using 127.0.0.1.
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000016s latency).
Other addresses for localhost (not scanned): ::1

PORT   STATE  SERVICE
22/tcp closed ssh
80/tcp closed http

Nmap done: 1 IP address (1 host up) scanned in 0.02 seconds
phanindra1815@Phanindras-PC:~$
```

4. lshw

```
phanindra1815@Phanindras-PC:~$ lshw -short
WARNING: you should run this program as super-user.
H/W path  Device  Class      Description
====================================
                  system     Computer
/0                bus        Motherboard
/0/0              memory     8064MiB System memory
/0/1              processor  11th Gen Intel(R) Core(TM) i5-11400H @ 2.70GHz
/0/2              storage    Virtio 1.0 console
/0/2/0            generic    Virtual I/O device
/0/3              display    Basic Render Driver
/0/4              generic    Virtio 1.0 file system
/0/4/0            generic    Virtual I/O device
/0/5              display    Basic Render Driver
/0/6              system     PnP device PNP0b00
/1        eth0    network    Ethernet interface
WARNING: output may be incomplete or inaccurate, you should run this program as super-user.
phanindra1815@Phanindras-PC:~$
```

5. ethtool

```
phanindra1815@Phanindras-PC:~$ ethtool eth0
Settings for eth0:
        Supported ports: [  ]
        Supported link modes:   Not reported
        Supported pause frame use: No
        Supports auto-negotiation: No
        Supported FEC modes: Not reported
        Advertised link modes:  Not reported
        Advertised pause frame use: No
        Advertised auto-negotiation: No
        Advertised FEC modes: Not reported
        Speed: 10000Mb/s
        Duplex: Full
        Auto-negotiation: off
        Port: Other
        PHYAD: 0
        Transceiver: internal
netlink error: Operation not permitted
        Current message level: 0x000000f7 (247)
                               drv probe link ifdown ifup rx_err tx_err
        Link detected: yes
phanindra1815@Phanindras-PC:~$
```

6. vmstat

```
phanindra1815@Phanindras-PC:~$ vmstat 2 5
procs -----------memory---------- ---swap-- -----io---- -system-- -------cpu-------
 r  b   swpd   free   buff  cache   si   so    bi    bo   in   cs us sy id wa st gu
 0  0      0 6478884  63772 990232    0    0  1271   545  883    0  0  0 99  0  0  0
 0  0      0 6477812  63772 990280    0    0     0     0  171  133  0  0 100  0  0  0
 0  0      0 6478132  63772 990280    0    0     0     0   45   27  0  0 100  0  0  0
 0  0      0 6479544  63780 990280    0    0     0     8  111   89  0  0 100  0  0  0
 0  0      0 6479544  63780 990280    0    0     0     0   61   32  0  0 100  0  0  0
phanindra1815@Phanindras-PC:~$
```

7. iostat

```
phanindra1815@Phanindras-PC:~$ iostat -xz 2 3
Linux 6.18.33.2-microsoft-standard-WSL2 (Phanindras-PC)         10/29/26        _x86_64_        (12 CPU)

avg-cpu:  %user   %nice %system %iowait  %steal   %idle
          0.28    0.00    0.30    0.11    0.00   99.31

Device            r/s     rkB/s   rrqm/s  %rrqm r_await rareq-sz     w/s     wkB/s   wrqm/s  %wrqm w_await wareq-sz     d/s     dkB/s   drqm/s  %drqm d_awai
t dareq-sz     f/s f_await  aqu-sz  %util
sda              1.66    329.87     0.91  35.44    0.45   198.98    0.00      0.00     0.00   0.00    0.00     0.00    0.00      0.00     0.00   0.00    0.0
0     0.00    0.00    0.00    0.00   0.04
sdb              0.23     13.46     0.14  38.55    0.40    58.92    0.00      0.00     0.00   0.00    0.00     0.00    0.00      0.00     0.00   0.00    0.0
0     0.00    0.00    0.00    0.00   0.01
sdc              0.20      4.09     0.00   0.00    0.06    20.94    0.00      0.01     0.00   0.00    2.00     2.00    0.00      0.00     0.00   0.00    0.0
0     0.00    0.00    2.00    0.00   0.00
sdd             20.64    964.09     4.93  19.27    0.62    46.71   11.16    503.47    29.85  72.78    2.07    45.11    1.54    149.53     0.05   3.14    0.1
2    97.27    6.28    0.68    0.04   1.04


avg-cpu:  %user   %nice %system %iowait  %steal   %idle
          0.00    0.00    0.12    0.00    0.00   99.88

Device            r/s     rkB/s   rrqm/s  %rrqm r_await rareq-sz     w/s     wkB/s   wrqm/s  %wrqm w_await wareq-sz     d/s     dkB/s   drqm/s  %drqm d_awai
t dareq-sz     f/s f_await  aqu-sz  %util


avg-cpu:  %user   %nice %system %iowait  %steal   %idle
          0.04    0.00    0.00    0.04    0.00   99.92

Device            r/s     rkB/s   rrqm/s  %rrqm r_await rareq-sz     w/s     wkB/s   wrqm/s  %wrqm w_await wareq-sz     d/s     dkB/s   drqm/s  %drqm d_awai
t dareq-sz     f/s f_await  aqu-sz  %util
sdd              0.00      0.00     0.00   0.00    0.00     0.00    1.00    114.00    27.50  96.49    2.00   114.00    2.50  28214.00     0.50  16.67    0.4
0 11285.60    1.00    1.00    0.00   0.20
```

8. fallocate

```
phanindra1815@Phanindras-PC:~$ fallocate -l 10M testfile
phanindra1815@Phanindras-PC:~$ ls
Hello.txt  Students.txt  directory  link.txt   log_part_aa  phani       sample.txt   snap
Linux.txt  backup        home       linux.txt  notes.txt    report.txt  sample.txt~  testfile
phanindra1815@Phanindras-PC:~$
```

9. getfacl

```
phanindra1815@Phanindras-PC:~$ getfacl sample.txt
# file: sample.txt
# owner: phanindra1815
# group: phanindra1815
user::rw-
group::r--
other::r--

phanindra1815@Phanindras-PC:~$
```

10. setfacl

```
phanindra1815@Phanindras-PC:~$ strace ls
execve("/usr/bin/ls", ["ls"], 0x7fff98a502b0 /* 25 vars */) = 0
brk(NULL)                               = 0x568bc39e5000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x739ad73a7000
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=21267, ...}) = 0
mmap(NULL, 21267, PROT_READ, MAP_PRIVATE, 3, 0) = 0x739ad73a1000
close(3)                                = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=199120, ...}) = 0
mmap(NULL, 206288, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x739ad736e000
mmap(0x739ad7375000, 135168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7000) = 0x739ad7375000
mmap(0x739ad7396000, 28672, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x739ad7396000
mmap(0x739ad739d000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2f000) = 0x739ad739d000
mmap(0x739ad739f000, 5584, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x739ad739f000
close(3)                                = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcc_s.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=187120, ...}) = 0
mmap(NULL, 185256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x739ad7340000
mmap(0x739ad7344000, 147456, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x4000) = 0x739ad7344000
mmap(0x739ad7368000, 16384, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x739ad7368000
mmap(0x739ad736c000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2c000) = 0x739ad736c000
close(3)                                = 0
```